

Demonstration of Proposed Features

Date	07 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures the core features proposed for OptiCrop during the project planning phase and records whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver the proposed machine-learning-based agricultural solution.

S. No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Soil & Climate Parameter Input	Accepts N, P, K, temperature, humidity, soil pH, and rainfall values	Implemented	Yes	Working Successfully
2	Input Validation	Validates pH, humidity, temperature, rainfall, and NPK values before prediction	Implemented	Yes	Working Successfully
3	ML Crop Recommendation	Uses the trained scikit-learn model to recommend a suitable crop	Implemented	Yes	Working Successfully
4	Prediction Confidence	Displays the model confidence percentage using class probabilities	Implemented	Yes	Working Successfully
5	Crop Guidance	Displays water requirement, suitable soil, fertilizer guidance, and cultivation tip	Implemented	Yes	Working Successfully

Feature Implementation Summary:

Total Features Proposed	5
Total Features Implemented	5
Total Features Demonstrated	5

Overall Implementation Rate (%)	100%
---------------------------------	------